

towards certain groups (liberals), but also pushes the model to almost embody caricatures of those groups (e.g., 99% approval of Joe Biden). From a different standpoint, this finding highlights the importance of considering the entire spectrum of human responses rather than just the mode. A modal analysis of text-davinci-003 would conclude that the model is highly representative of Democrats, where in reality its representation collapses the diversity of opinions held by different democrats into a single, modal response.

Refusals. In our comparison of human and LM opinions so far, we omitted the “refusal” option for all questions due to its non-ordinal nature. In Appendix B.1, we thus separately compare the refusal rates of LMs and human respondents. We find that all models have low refusal rates. Although human feedback-tuned models are encouraged to refuse to take a stance on contentious issues (Askell et al., 2021; Ouyang et al., 2022), they tend to rarely do so in our multiple-choice setting—with refusal rates as low as 1–2%.

4.2 Steerability

We now shift our focus from measuring the default alignment of LM opinions with those of various demographics groups *without* prompting, to studying their steerability *with* group-specific prompting. This is especially important in settings such as personalization, where a key measure of performance is an LM’s ability to adapt to represent the opinion of various demographic groups.

The metric. We measure steerability as the average opinion alignment, across dataset questions, between an LM and a particular demographic group G – where the model is prompted with group information in its context. Since our goal is to test whether a model *can* be steered toward a group, we consider three prompting strategies—QA, BIO, PORTRAY (see Section 3.1)—for each question and choose the one that works best. Concretely, we measure steerability as:

$$\mathcal{S}_m^G(Q) = \frac{1}{|Q|} \sum_{q \in Q} \max_{c_G \in [\text{QA}, \text{BIO}, \text{POR}]} \mathcal{A}(D_m(q; c_G), D_G(q))$$

where $D_m(q; c_G)$ denotes the LM opinion distribution conditioned on the group-specific context c_G . A higher $\mathcal{S}_m^G$ score indicates that the model is better aligned to the opinions of the given group. Note that unlike default subgroup representativeness, an LM’s steerability could be simultaneously high for multiple (disagreeing) groups. In fact, in many cases, we might want disparities in the default subgroup representativeness scores of an LM to be remedied by steering.

Steering does not solve opinion misalignment. We attempt to steer current LMs towards one of 22 demographic groups (e.g., Republican, Asian, Jewish) listed in Appendix Table 4 on a subset Q_S of 500 highly contentious questions from OpinionQA. In Figure 4b, we compare different LMs in terms of their ability to match the opinions of a subgroup on these questions, by default and with steering ($\mathcal{S}_m^G(Q_S)$ from Section 4.1).

Most LMs (with the exception of ada) do become somewhat more representative of a subpopulation post-steering. However, none of the disparities in group opinion alignment of an LM disappear after steering, with text-davinci-002 showing the smallest post-steering alignment gap across groups. In most cases, we see the representativeness of all groups improving by a constant factor—indicating that the LM still does better on some groups than others. In Appendix Figure 11, we visualize which LMs are most effective at adapting towards a particular group: e.g., j1-grande-v2-beta for Southerners and text-davinci-002 for liberals.

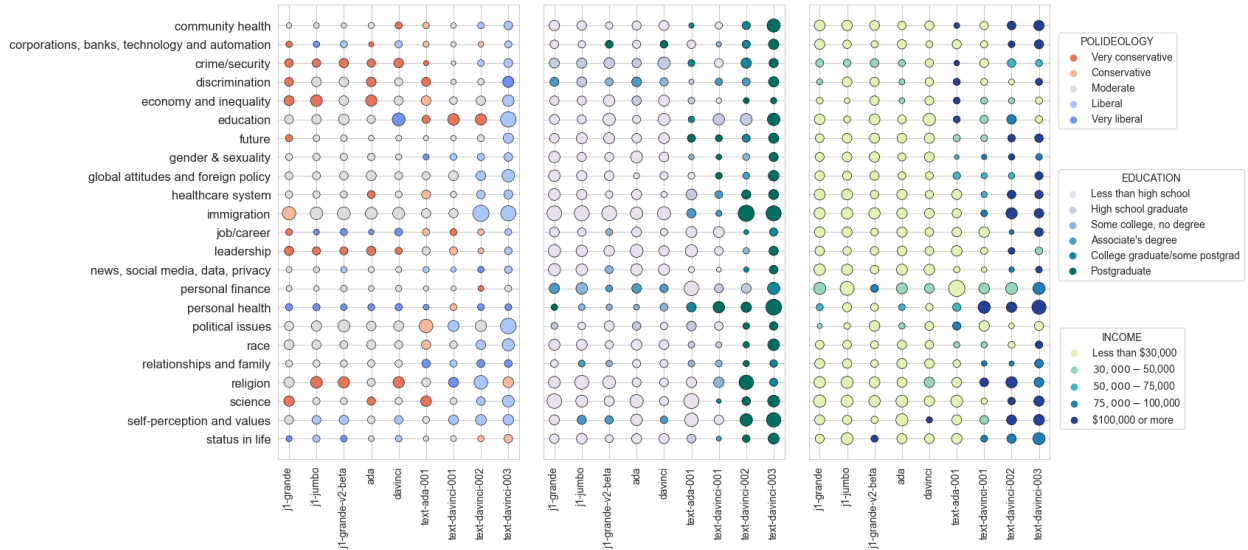


Figure 5: Consistency of different LMs (columns) across topics (rows) on different demographic attributes (panels). Each dot indicates an LM-topic pair, with the color indicating the group to which the model is best aligned, and the size of the dot indicates the strength of this alignment (computed as the ratio of the best and worst subgroup representativeness for that topic, see Appendix B.3 for details). We find significant topic-level inconsistencies, especially for base LMs, and strong educational attainment consistency for RLHF trained LMs.

4.3 Consistency

Our earlier default representativeness analysis (Section 4.1) showed marked skews in the views expressed by LMs, with base LMs reflecting opinions consistent with lower income and education and the opposite for human-feedback tuned ones. However, we might want to go beyond this aggregate analysis and ask: *are the views expressed by LMs consistent across topics?* (Saris & Sniderman, 2004). For instance, is `text-davinci-002` politically Liberal on all matters or does it take a Conservative stance in some cases? We now leverage the fine-grained topic taxonomy in our OpinionQA dataset to answer this question. To this end, we inspect human-LM opinion similarity on a topic level by computing alignment on a subset of questions Q_T .

Are LMs consistent? In Figure 5, we break down the subgroups that various LMs (columns) most closely align to (colors) across 23 topic categories (rows) by political ideology, education and income. Of all the LMs we study, the base models from both providers and the RLHF-trained `text-davinci-003` from OpenAI seem to be the most consistent – albeit towards different sets of groups. None of the models are perfectly consistent however, and even `text-davinci-00{2,3}` aligns with conservatives on topics like religion.

The metric. To distill these trends into a single measure, we ask what is the fraction of topics for which an LM’s most aligned group overall (weighting topics equally) matches the LM’s most aligned group on the given topic (with questions Q_i). Specifically, for a model, we first identify the

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j1-grande	j1-jumbo	j1-grande-v2-beta	ada	davinci	text-ada-001	text-davinci-001	text-davinci-002	text-davinci-003
0.612	0.612	0.575	0.622	0.562	0.388	0.405	0.502	0.575

Figure 6: Consistency of LM opinions $\mathcal{C}_m$, where a higher score (lighter) indicates that an LM aligns with the same set of groups across topics.

group it best aligns to across topics as

$$G_m^{best} := \arg \max_G \left(\frac{1}{T} \sum_{T'} \mathcal{R}_M^G(Q_{T'}) \right)$$

We then define consistency as:

$$\mathcal{C}_m := \frac{1}{T} \sum_T \mathbb{1} \left[\left(\arg \max_G \mathcal{R}_M^G(Q_T) \right) = G_m^{best} \right]$$

Our metric $\mathcal{C}_m$ is bounded between 0 and 1, and a higher score implies that the model agrees with the views of the same subgroups across all topics. In Figure 6, we visualize the average consistency score of a model across demographic traits (religion/income/ideology, etc). We find that the overall consistency scores of current LMs are fairly low—indicating that they are expressing a patchwork of disparate opinions. Note that this may not always be problematic—after all even individuals can hold seemingly inconsistent beliefs.

5 Related work

Evaluating LM personas. There has been growing interest in probing the ability of LMs to mimic human behaviors. One line of work asks whether LMs can replicate results from well-known human experiments, e.g., in cognitive science, social science, and economics (Uchendu et al., 2021; Karra et al., 2022; Aher et al., 2022; Binz & Schulz, 2022; Srivastava et al., 2022). Another set of studies have examined whether LMs can be used to simulate personas (Park et al., 2022; Argyle et al., 2022; Jiang et al., 2022; Simmons, 2022), akin to our notion of steerability. Through case studies in specific settings, these works gauge whether prompting LMs with demographic information (e.g., political identity) leads to human-like responses: Argyle et al. (2022) look at voting patterns and word associations, and Simmons (2022) consider moral biases. By leveraging public opinion surveys, we are able to improve our understanding of LM steerability in three ways: (i) *breadth*: both in the range of different topics and steering groups, (ii) *distributional view*: gauging whether LMs can match the spectrum of opinions of a group rather than its modal opinion, and (iii) *measurability*: using metrics grounded in human response distributions.

Finally, recent works have examined the slants in the opinions of LMs—by prompting them with contentious propositions/questions generated by LMs Perez et al. (2022b) or from political tests Hartmann et al. (2023). Similar to our work, they find that human-feedback trained models often exhibit a left-leaning, pro-environmental stance. However, since our approach is based on public opinion surveys, we can go beyond the modal perspective taken by these works (comparing models to dominant viewpoints of specific groups, e.g., pro-immigration for liberals). Instead, we